

Cascading Style Sheets(CSS)

Cascading Style Sheets(CSS)

- Used to style a web page
- It is a style sheet language which is used to describe the look and formatting of a document written in markup language
- Provides an additional feature to HTML. Generally used with HTML to change the style of web pages and user interfaces
- Using CSS we can control the layout of multiple pages at once

Why CSS?

- Saves time
 - Using external CSS we can control the layout of multiple pages at once
- More attributes
 - CSS provides more detailed attributes than plain HTML to define the look and feel of the website
- Solves repeated use of certain attributes
 - Without CSS, tags like font, color, background style, element alignments, border and size had to be repeated on every web page which was a tedious process.
 - Ex. While developing a large website where fonts and color information are added on every single page, it will become a long and expensive process

CSS Syntax

- A CSS rule set consist of a **Selector** and a **Declaration block**

```
h1{ color: yellow; font-size: 11 px;}
```

- **Selector:** Indicates the HTML element you want to style. It could be any tag like <h1>, <title> etc.
- **Declaration Block:** Contains one or more declarations separated by a semicolon. The above example, consists of two declarations - **color: yellow;** and **font-size: 11 px;**
 - Each declaration contains a **Property name** and **Value**, separated by a colon

Types of CSS

There are three ways to insert CSS in HTML documents

- **Inline CSS**

- To apply CSS on a **single line or an element**
- **Style attribute** is added to the relevant tag

```
<p style="color:blue">Hello</p>
```

Types of CSS

- Internal or Embedded CSS

- Used to add a unique style for a single document
- It is defined in the <head> section of the HTML page inside the <style> tag

```
<html>
<head>
<style>
h1 {
  color: red;
  margin-left: 80px; }
</style>
</head>
<body>
<h1>The internal style sheet is applied on this heading.</h1>
</body>
</html>
```

Types of CSS

- External CSS

- Used when changes have to be made on multiple pages
- Helps to change the look of the entire web site by changing just one file
- It uses the <link> tag on every pages and the <link> tag should be put inside the head section

```
<head>  
<link rel="stylesheet" type="text/css" href="styl1.css">  
</head>
```

Types of CSS

styl1.css

```
body {  
    background-color: lightblue;  
}  
  
h1 {  
    color: navy;  
    margin-left: 20px;  
}
```


CSS Comments

- Comments are single or multiple lines statement written within `/*.....*/`
- Comments are ignored by browsers

CSS Selectors

- Used to **select the content you want to style**. CSS selectors select HTML elements according to its **id, class, type, attribute** etc.
- Different types of selectors in CSS are
 - CSS Element Selector**
 - CSS Id Selector**
 - CSS Class Selector**
 - CSS Universal Selector**
 - CSS Group Selector**

CSS Element Selector

- Selects the html element by name

```
<html>
<head>
<style>
p{
    text-align: center;
    color: blue;
}
</style>
</head>
<body>
<p>MESCE</p>
<p>MCA</p>
</body>
</html>
```

CSS id Selector

- Selects the id attribute of an HTML element to select a specific element
- An id is always unique within the page so it is chosen to select a single, unique element.
- It is written with the hash character (#), followed by the id of the element

```
<html>
<head>
<style>
#p1{
    text-align: center;
    color: blue;
}
</style>
</head>
<body>
<p>MESCE</p>
<p id = "p1">MCA</p>
</body>
</html>
```

CSS Class Selector

- The class selector selects HTML elements with a specific class attribute
- Used with a period character (.) followed by the class name

```
<html>
<head>
<style>
.center {
    text-align: center;
    color: blue;
}
</style>
</head>
<body>
<h1 class="center">MESCE</h1>

<p class="center">MCA</p>
</body>
</html>
```

CSS Class Selector for Specific Element

```
<html>
<head>
<style>
p.center {
    text-align: center;
    color: blue;
}
</style>
</head>
<body>
<h1 class="center">MESCE</h1>
<p class="center">MCA</p>
</body>
</html>
```

—————→ *Element name with the selector*

CSS Universal Selector

- The universal selector is used to select all the elements in a html page
- Denoted by an asterisk character

```
<html>
<head>
<style>
* {
    font-size: 20px;
    color: green;
}
</style>
</head>
<body>
<h1>MESCE</h1>
<p>MCA</p>
</body>
</html>
```

CSS Group Selector

- Used to select multiple elements and style them together
- Reduces the code and effort to declare styles for each element

```
<html>
<head>
<style>
h1, p {
    text-align: center;
    color: green;
}
</style>
</head>
<body>
<h1>MESCE</h1>
<p>MCA</p>
</body>
</html>
```


